



HR Overtime Management System

Technical & Functional Documentation — Odoo 19

Modules: hr_overtime_management + hr_overtime_payroll (optional)

Version: 19.0.1.4.0

Document date: July 06, 2026

Database: odoo19



1. Executive Summary

This document describes the custom Odoo 19 overtime management solution delivered for your organization. The system allows employees to submit overtime requests linked to projects and tasks, routes them through a multi-level approval chain (department manager → optional upper manager → HR), calculates overtime cost from the employee contract wage, optionally creates timesheet lines on final approval, and optionally pushes costs to payroll.

The implementation follows Odoo best practices: reusable approval engine (mirroring hr_holidays patterns), configurable pay-rate multipliers, no duplicate time-tracking tables, and optional payroll integration in a separate glue module.

2. What Was Customized vs What Already Existed in Odoo

This section separates deliverables built for your organization from standard Odoo features that were reused without re-implementing them.

2.1 Already Existed in Odoo (Reused)

Odoo Standard	How We Use It
hr.employee + parent_id chain	Manager / upper-manager resolution (same idea as Time Off)
hr.version (contract wage)	Hourly cost from employee wage and work schedule
project.project / project.task	Project & task on each overtime request
account.analytic.line (hr_timesheet)	Optional timesheet line on final approval
resource.calendar.leaves	Public holidays / calendar days off for auto type detection
mail.thread / mail.activity	Chatter, notifications, approval activities
res.company multi-company	Company branches, record rules, allowed companies
hr_payroll (optional)	Payslip input via hr_overtime_payroll glue module only
Security groups (res.groups)	Standard privilege / group pattern

2.2 Custom-Built for This Project

Custom Component	Purpose
hr.overtime.request	Main overtime request with workflow states
hr.overtime.type	Per-branch rate categories: Regular, Weekend, Day Off
hr.overtime.approval.line	Audit trail for each approval step
hr.approval.chain.service	Reusable approval-chain resolver (manager → HR)
hr.approval.chain.mixin	Reusable submit / approve / refuse workflow
Auto overtime type by date	Picks Regular / Weekend / Day Off from datetime + calendar
Per-company type provisioning	Each company branch gets its own 3 types + multipliers
Middle-East weekend (Fri/Sat)	Configurable weekend weekdays per company
Multi-company employee access	Employees only see projects / companies they belong to
overtime_error_handler.js	Friendly snackbar for company / access errors



QWeb PDF report	Printable overtime approval document
Refuse wizard	Mandatory reason when refusing a request

2.3 Customizations Applied During Delivery

- Datetime pickers (start_datetime / end_datetime) instead of separate date + float times.
- Overtime Type is auto-computed — employees do not manually select the category.
- Three category enums on hr.overtime.type: regular, weekend, day_off.
- Each company branch has its own overtime types with independent rate multipliers.
- New companies auto-receive Regular / Weekend / Day Off types on creation.
- Total cost = hours × hourly wage × rate_multiplier (multiplier editable per branch).
- Top-level Overtime app menu (My Requests, My Approvals, All Requests, Configuration).
- Server actions for My Requests / My Approvals with correct multi-company context.

3. Business Need & Objectives

The following business requirements drove this development:

- Employees submit overtime with project/task allocation and supporting documents.
- Approval follows the existing organizational hierarchy (manager chain + mandatory HR).
- Overtime pay rates per branch (Regular / Weekend / Day Off) are configurable without code.
- Approved overtime can generate account.analytic.line (timesheet) entries automatically.
- Overtime cost is derived from contract wage, not a parallel payroll engine.
- A reusable approval-chain component supports future request types (expenses, permissions, etc.).
- Security: employees see own requests; managers see pending approvals; HR sees all.

4. Delivered Modules

Module	Type	Purpose	Dependencies
hr_overtime_management	Core (required)	Requests, approval workflow, reporting, settings	hr, hr_timesheet, project, mail, resource
hr_overtime_payroll	Optional glue	Push approved overtime to payslip inputs	hr_overtime_management, hr_payroll

5. System Architecture



5.1 Reusable Approval Engine

Component	Model	Role
Chain Service	hr.approval.chain.service	Resolves manager → upper manager → HR chain from employee.parent_id
Chain Mixin	hr.approval.chain.mixin	Generic submit / approve / refuse workflow with mail activities
Overtime Request	hr.overtime.request	Concrete implementation using the mixin

5.2 Approval Chain Logic

Mirrors Odoo hr_holidays manager resolution pattern:

- Step 1: employee.parent_id (or department manager if no parent) → dept_manager
- Step 2: parent of dept manager (if different person) → upper_manager
- Step 3: HR officer from group_overtime_hr_officer → hr (always mandatory final stage)

Two scenarios:

- 2 stages: Solo manager → dept_manager → HR
- 3 stages: dept_manager → upper_manager → HR

5.3 Workflow States

State	Meaning
draft	Employee is editing; not yet submitted
submitted	Waiting on first approver (dept manager)
manager_approved	Dept manager approved; waiting upper manager or HR
upper_manager_approved	Upper manager approved; waiting HR
hr_approved	Fully approved — triggers timesheet / payroll hooks
refused	Rejected at any stage (reason required)
cancel	Cancelled by employee before completion

5.4 Automatic Overtime Type Selection

Overtime type is computed from the request period and the employee company branch:

Category (enum)	When Applied	Default Multiplier
regular	Normal working days (Sun–Thu by default)	1.5×
weekend	Configured weekend weekdays (default Fri=4, Sat=5)	2.0×
day_off	Public holiday on employee working calendar	2.5×



If the period spans multiple categories, the highest multiplier wins. Each company branch maintains its own hr.overtime.type records — San Francisco and Chicago can use different multipliers.

6. Database Models & Tables

PostgreSQL tables created or extended by hr_overtime_management. Standard Odoo tables (hr_employee, project_project, etc.) are not duplicated.

6.1 hr.overtime.type → table hr_overtime_type

Column	Type	Description
id	Integer	Primary key
name	Varchar	Display name (translatable)
code	Varchar	Technical code: regular, weekend, holiday
category	Varchar	Enum: regular weekend day_off
rate_multiplier	Float	Pay multiplier — editable per branch (e.g. 1.5, 2.0, 2.5)
sequence	Integer	Display order in lists
company_id	Integer FK → res_company	Company branch (NULL = shared template)
active	Boolean	Archive inactive types
create_uid / write_uid	Integer	Audit — Odoo standard
create_date / write_date	Timestamp	Audit — Odoo standard

6.2 hr.overtime.request → table hr_overtime_request

Column	Type	Description
id	Integer	Primary key
name	Varchar	Sequence OT/YYYY/MM/00001
employee_id	Integer FK → hr_employee	Employee who worked overtime
department_id	Integer FK	Stored related from employee
manager_id	Integer FK	Direct manager (parent_id)
employee_company_id	Integer	Employee company id (computed, stored)
start_datetime	Timestamp	Overtime start
end_datetime	Timestamp	Overtime end
date	Date	Computed from start_datetime
overtime_hours	Float	Computed duration in hours
overtime_type_id	Integer FK → hr_overtime_type	Auto-computed from date/category
project_id	Integer FK → project_project	Timesheet-enabled project
task_id	Integer FK → project_task	Task within project
description	Text	Reason / details
state	Varchar	draft submitted manager_approved upper_manager_approved hr_approved refused cancel
current_approver_id	Integer FK → res_users	User who must act now
hourly_cost	Numeric	Computed from hr.version wage
total_cost	Numeric	hours × hourly_cost × rate_multiplier



analytic_line_id	Integer FK	Timesheet line after HR approval
company_id	Integer FK → res_company	Employee company branch
currency_id	Integer FK	Company currency

Many2many hr_overtime_request_ir_attachment_rel links supporting documents (ir_attachment).

6.3 hr.overtime.approval.line → table hr_overtime_approval_line

Column	Type	Description
request_id	Many2one	Parent overtime request
sequence	Integer	Order in chain (10, 20, 30...)
role	Selection	dept_manager upper_manager hr
approver_id	Many2one	Assigned approver user
state	Selection	pending to_approve approved refused
decision_date	Datetime	When decision was made
comment	Text	Refusal reason or notes

6.4 hr.approval.chain.service (Abstract — no table)

AbstractModel — logic only. Resolves (role, approver_user) tuples for any employee.

6.5 hr.approval.chain.mixin (Abstract — no table)

AbstractModel — generic submit, approve, refuse, activity scheduling.

6.6 res.company.extensions (columns added to res_company)

Column	Type	Default	Description
overtime_generate_analytic_line	Boolean	True	Auto-create timesheet on HR approval
overtime_default_type_id	Integer FK	Regular type	Regular working day multiplier
overtime_weekend_type_id	Integer FK	Weekend type	Weekend day multiplier
overtime_holiday_type_id	Integer FK	Day Off type	Public holiday multiplier
overtime_weekend_weekdays	Varchar	4,5	Weekend weekday numbers (Mon=0)
overtime_daily_hours_cap	Float	4.0	Warning threshold for long requests
overtime_hours_per_month	Float	173.33	Wage ÷ hours fallback for hourly cost

6.7 Entity Relationship Summary

hr_employee → hr_overtime_request → hr_overtime_approval_line

hr_overtime_request → hr_overtime_type (by category + company branch)



hr_overtime_request → project_project / project_task → account_analytic_line (optional)

res_company → 3× hr_overtime_type (regular, weekend, day_off) per branch

7. Cost Calculation

Formula: total_cost = overtime_hours × hourly_cost × overtime_type.rate_multiplier

7.1 Hourly Cost (Odoo 19)

In Odoo 19, contract data lives on hr.version (linked via employee.current_version_id). Hourly cost is calculated as:

- Primary: wage × 12 ÷ 52 ÷ hours_per_week (from employee work schedule)
- Fallback: wage ÷ overtime_hours_per_month (company setting, default 173.33)

Example — Administrator, wage \$7,540/month, 38 h/week, 2 h Regular OT (1.5×): Hourly ≈ \$45.81 → Total ≈ \$137.43

8. Security Groups & Access

Group	Users	Access
group_overtime_user	All employees	Create/read own requests; edit only in draft
Managers (record rule)	current_approver_id = me	Approve/refuse pending requests
group_overtime_hr_officer	HR team	Read all; final HR approval
group_overtime_admin	Administrators	Configure types, settings, full access

9. User Interface

Menu	Purpose
Overtime → My Requests	Employee self-service list/form
Overtime → My Approvals	Kanban for pending approvers
Overtime → All Requests	HR officers — full list
Overtime → Configuration → Overtime Types	Per-branch Regular / Weekend / Day Off multipliers
Settings → Employees → Overtime	Company branch settings + type links

Views delivered: List, Form (statusbar + approval tab + chatter), Kanban, Search filters, Pivot, Graph, QWeb PDF report.

10. Integrations

10.1 Timesheets (account.analytic.line)



When overtime_generate_analytic_line is enabled (default), final HR approval creates a timesheet line with project, task, employee, date, and overtime_hours as unit_amount.

10.2 Payroll (optional — hr_overtime_payroll)

If hr_payroll is installed and overtime_link_to_payroll is enabled, HR approval creates/updates an hr.payslip.input line (type Overtime) on the employee's open payslip.

11. Issues Encountered & Resolutions

Issue	Cause	Resolution
Submit crash: res.groups has no attribute users	Odoo 19 uses all_user_ids not users	Fixed get_hr_responsible() to use all_user_ids
column start_datetime does not exist	Code updated but module not upgraded on odoo19	SQL migration + module upgrade
OwlError: start_time field undefined	Stale ir.ui.view records in database	Updated stored view XML in odoo19
Approval chain incomplete: no HR stage	Sanitizer removed HR when same user as manager	HR stage always preserved in chain
Hourly/Total cost \$0.00	Core returned 0; Odoo 19 uses hr.version not hr.contract	Read wage from employee.version_id
column overtime_hours_per_month missing	New field without DB upgrade	Added column + reset module state
Missing overtime_type_id on save	Computed field not set before INSERT	Pre-resolve type in create(); category enums
overtime_weekend_weekdays column missing	Module not upgraded	SQL migration + module upgrade
Menu server action TypeError	action context is a string not dict	literal_eval before merge
Multi-company project access	Session company ≠ employee company	allowed_company_ids + record rules

12. Configuration Checklist (Multi-Company)

- 1. Install hr_overtime_management on database odoo19.
- 2. For each company branch: verify 3 overtime types exist (auto-created on company create).
- 3. Set rate multipliers per branch under Overtime → Configuration → Overtime Types.
- 4. Configure weekend weekdays per branch in Settings → HR → Overtime (default 4,5 = Fri/Sat).
- 5. Assign users to Officer: Overtime HR Approval for each branch that needs HR approval.
- 6. Ensure employees have wage on hr.version and projects allow timesheets.
- 7. Optional: install hr_overtime_payroll for payslip integration.

13. End-User Guide



13.1 Employee — Submit Overtime

- Go to Overtime → My Requests → New.
- Set start/end date & time, project, task, description (type is auto-set from date).
- Attach documents if needed → Save → Submit.
- Overtime Type and Total Cost update automatically (Regular / Weekend / Day Off).

13.2 Manager — Approve

- Go to Overtime → My Approvals.
- Open pending request → Approve or Refuse (reason required).
- If you are also HR, you may need to approve twice (manager step then HR step).

13.3 HR — Configure Branch Rates

- Switch to the company branch (company selector in top bar).
- Open Overtime → Configuration → Overtime Types — grouped by company.
- Edit Regular / Weekend / Day Off records and change Rate Multiplier per branch.
- Or use Settings → HR → Overtime to link types and set weekend weekdays.

13.4 HR — Final Approval

- Any user in Officer: Overtime HR Approval can perform final approval.
- On approval: state → Approved; timesheet line created if enabled; payroll input if enabled.

14. Technical File Structure

extra_addons/hr_overtime_management/

- models/hr_approval_chain_service.py — reusable chain resolver
- models/hr_approval_mixin.py — generic workflow mixin
- models/hr_overtime_request.py — main request + state machine
- models/hr_overtime_approval_line.py — approval trail
- models/hr_overtime_type.py — configurable rate types
- security/security.xml — groups + record rules
- views/ — list, form, kanban, search, pivot, graph
- report/hr_overtime_report.xml — QWeb PDF



- wizard/hr_overtime_refuse_wizard.py — refusal reason dialog
- hooks.py — post_init: provision overtime types for all company branches
- tests/test_hr_overtime.py — automated tests
- doc/generate_full_documentation.py — this PDF/DOCX generator

15. Upgrade & Maintenance

After any code change, upgrade the module:

```
python odoo-bin -c debian/odoo.conf -d odoo19 -u hr_overtime_management --stop-after-init
```

Or: Apps → HR Overtime Management → Upgrade

Always use database odoo19 (not odoo) for this installation.

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